

# Answers 1 and 2

Open after attempting the test. Award method marks for a correct setup with arithmetic errors.

## 1. Equilibrium [10]

(a) Weight  $mg$  down, tension  $T$  along the string toward the support, electric repulsion horizontally away from the other sphere. [2]

(b)  $T \cos(\theta) = mg$ ;  $T \sin(\theta) = F_e$ , so  $F_e = mg \tan(\theta)$ . [2]

$r = 2L \sin(\theta) = 1.014 \text{ m}$ . [1]

$|q| = r \sqrt{mg \tan(\theta)/k} = 2.80 \times 10^{-6} \text{ C}$ . [2]

Each charge is **-2.80 microcoulombs**. [1]

(c) The angle increases: shorter strings reduce the separation at the old angle, increasing repulsion. A larger angle is needed to restore balance. [2]

## 2. Flux [10]

(a) Use  $\Phi = Q_{\text{enclosed}}/\epsilon_0$ :

$S_A$ :  $+452 \text{ N m}^2/\text{C}$ ;

$S_B$ :  $-429 \text{ N m}^2/\text{C}$ ;

$S_C$ :  $-158 \text{ N m}^2/\text{C}$ . [1 each]

(b) No change in net flux: enclosed charge is unchanged. The local field on the surface can change. [2]

(c)  $A = 0.0100 \text{ m}^2$ ;  $E_x = E \cos(53.1 \text{ degrees})$ ,  $E_y = E \sin(53.1 \text{ degrees})$ .

$+x$ :  $+24.0$ ;  $-x$ :  $-24.0$ ;

$+y$ :  $+32.0$ ;  $-y$ :  $-32.0$ ;

$+z$  and  $-z$ :  $0$ .

All flux values are in  $\text{N m}^2/\text{C}$ . Total flux =  $0$ ; net enclosed charge =  $0$ . [3 for face fluxes, 1 for total, 1 for enclosed charge]

**Watch for:** using  $L$  instead of the full sphere separation; dropping charge signs in Gauss's law; treating inward and outward flux as both positive.

# Answers 3 and 4

Derivation and maximum-field calculation

## 3. Ring derivation [10]

(a) Let  $dq = \lambda dl$ , with  $\lambda = Q/(2\pi R)$ . Each element is at distance  $r = \sqrt{x^2 + R^2}$  from P. [2]

(b) Opposite ring elements cancel transverse components. Axial components add toward +x. [2]

(c)  $dE_x = (k dq/r^2)(x/r)$ . Therefore

$$E_x = kx/(x^2 + R^2)^{3/2} \text{ times the integral of } dq$$

$$= kQx/(x^2 + R^2)^{3/2}. [4]$$

(d) At  $x = 0$ ,  $E = 0$  by symmetry. For  $x$  much larger than  $R$ ,  $E$  approaches  $kQ/x^2$ , the point-charge result. [2]

## 4. Maximum [10]

(a) Differentiate the product  $x(x^2 + R^2)^{-3/2}$ :

$$dE/dx = kQ[(x^2 + R^2)^{-3/2} - 3x^2(x^2 + R^2)^{-5/2}]$$

$$dE/dx = kQ(R^2 - 2x^2)/(x^2 + R^2)^{5/2}. [4]$$

(b)  $R^2 - 2x^2 = 0$  gives  $x = R/\sqrt{2}$ . [2]

(c) The derivative is positive before this point and negative afterwards.  $E$  is zero at  $x = 0$  and tends to zero at infinity. [2]

(d) Substitute  $x = R/\sqrt{2}$ :

$$E_{\max} = 2kQ/(3\sqrt{3} R^2). [2]$$

**Watch for:** forgetting the axial projection  $x/r$  in the derivation; differentiating  $Q$  or  $R$  as though they vary with  $x$ ; reporting only the position when the maximum field value is also requested.

## Bonus answer

Non-uniform rod: setup first, integration second

### Bonus 1 [5]

$$dq = \lambda_0 (1 + x/L) dx. [1]$$

The source-to-point distance is  $x + d$ . [1]

Every element produces a field toward  $-x$  at P. [1]

$$E_x = -k \lambda_0 \int_0^L \frac{(1 + x/L)}{(x + d)^2} dx. [2 \text{ for kernel and limits}]$$

### Bonus 2 [5]

Let  $u = x + d$ , so  $1 + x/L = (1 - d/L) + u/L$ .

An antiderivative is:

$$-(1 - d/L)/(x + d) + (1/L) \ln(x + d). [2]$$

After evaluating the bounds:

$$E_x = -k \lambda_0 \left\{ (1 - d/L) \left[ \frac{1}{d} - \frac{1}{L + d} \right] + \frac{1}{L} \ln \left[ \frac{L + d}{d} \right] \right\}. [1]$$

For  $L = 0.500$  m and  $d = 0.100$  m, the bracket is  $10.2502 \text{ m}^{-1}$ .

$$E_x = -184 \text{ N/C}, \text{ i.e. magnitude } 184 \text{ N/C toward the left. [2]}$$

### Quick self-check

Main score: \_\_\_\_\_ / 40    Bonus: \_\_\_\_\_ / 10

Circle the skill to revisit: force diagrams / flux signs / derivations / differentiation / integration.

Redo your weakest question without the key. Explain the step you missed before trying another example.

The mock's timing, mark allocations, chosen ring function and bonus variant are study-design choices. The instructor's announcement confirms the topic categories, not these exact questions.